higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL CONSTRUCTION N6

(8060026)

20 November 2019 (X-Paper)
09:00–13:00

REQUIREMENTS: Hot-rolled structural steel sections (Red Book) (BOE 8/6)

Personal notes and textbooks may be used.

This question paper consists of 7 pages and 3 schedules.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL CONSTRUCTION N6
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in the question paper.
4. Start each question on a NEW page.
5. Draw a line across the page at the end of each answer.
6. ALL calculations must conform to the relevant SABS/SANS Codes of Practice.
7. Indicate ALL relevant code/clause references.
8. Complement answers with neat sketches where applicable.
9. Use the SCHEDULES A, B and C (attached) to assist with answers.
10. Write neatly and legibly.

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QUESTION 1

FIGURE 1 shows a connection at the tie beam (bottom boom) of a steel roof truss where TWO discontinuous rolled steel angles are connected to the tie beam by means of an 8 mm thick gusset plate.

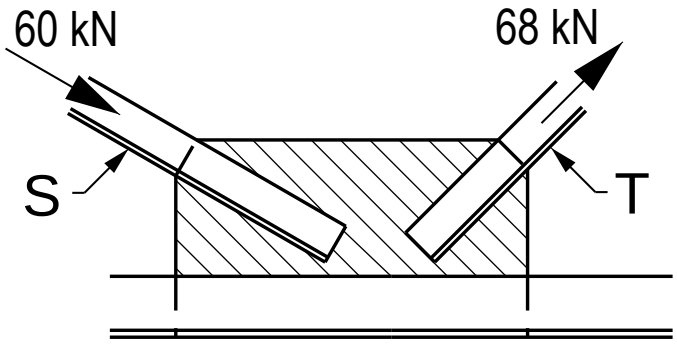


FIGURE 1

1.1 The part marked T is bolted to the gusset plate with grade 4,6 M16 bolts.

Determine the number of bolts required to secure the part T to the gusset safely. The thread of the bolts will be in the shearing plane. (6)

1.2 Select a suitable equal-leg angle section for the part marked S. The total length of part P is 1,48 m. (8)

[14]

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QUESTION 2

FIGURE 2 shows an eccentrically loaded connection consisting of a plate connected to the flange of an I-beam with FOUR bolts. The connection is required to support a load of 18 kN.

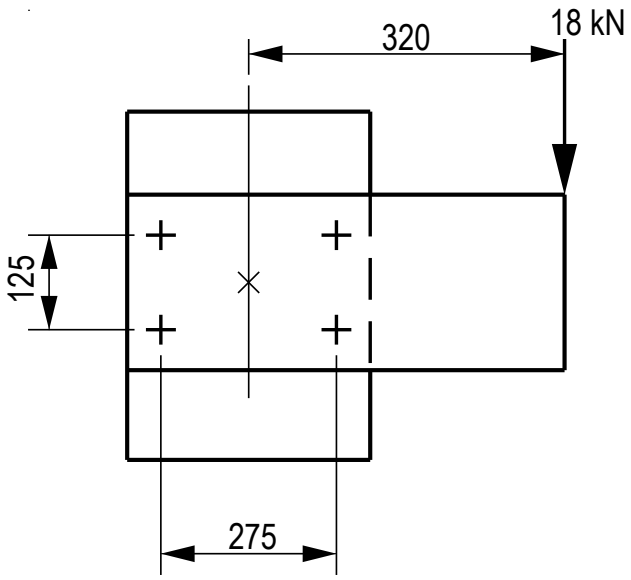


FIGURE 2

Calculate the diameter of the bolts needed to hold the connection in place. Use a shear stress of 90 MPa. **[15]**

QUESTION 3

FIGURE 3 shows a one-directional, simply supported, reinforced solid concrete slab to be cast across a dry river bed to form a pedestrian walkway. The slab will be supported on TWO I-parallel flange steel beams. The walkway must be able to support a live load of 8 kN/m.

The specifications of the slab are as follows:

Effective span:	4 m
Length of the slab:	8 m
Overall depth:	0,30 m
Density of concrete:	2 400 kg/m³

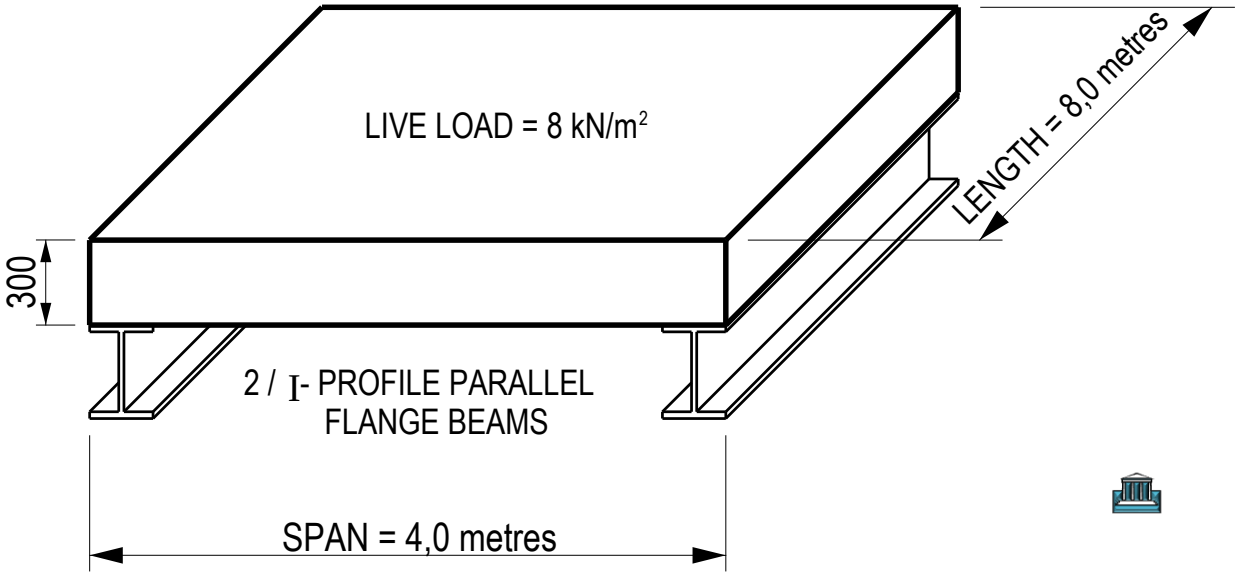


FIGURE 3

Calculate:

3.1 The total design dead load of the solid slab (5)

3.2 A suitable I-parallel flange steel beam to support the slab at A (7)

[12]

QUESTION 4

FIGURE 4 shows an I-section parallel flange steel beam with an effective span of 7,0 metres. The beam is simply supported over TWO reactions and is required to support the TWO given loads as well as its own weight.

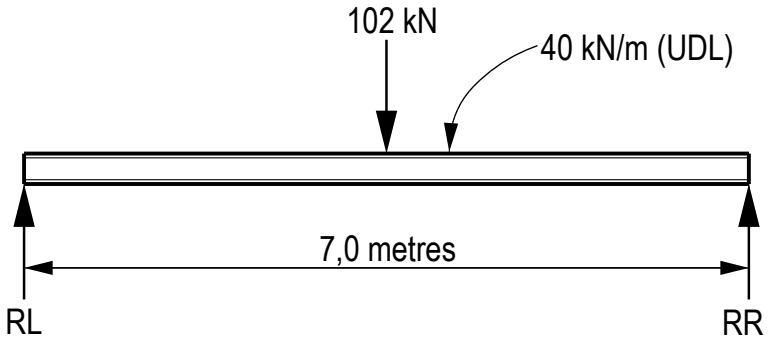


FIGURE 4

4.1 Use a bending stress of 163 MPa to calculate and select a suitable I-parallel flange steel beam to support the given loads. (7)

4.2 Include the self-weight of the selected steel beam then check if the beam is adequate to carry the imposed loads as well as its own self-weight. (8)
[15]

QUESTION 5

A simply supported rectangular beam must be cast over a double garage door opening. The effective span of the beam is 5,5 metres and it must support an imposed load of 3,5 kN/m. The width of the beam is 270 mm. Assume the density of concrete to be 2 400 kg/m³. Consider the self-weight of the reinforced concrete beam.

Use grade 25 concrete with mild-steel reinforcement and determine the following:

5.1 The effective depth of the reinforced concrete beam (14)

5.2 The suitable tension reinforcement for the reinforced concrete beam (2)

5.3 The minimum and maximum required main reinforcement (4)
[20]

QUESTION 6

A short, round, axially loaded reinforced concrete column with a diameter of 525 mm has to support an ultimate load of 3 220 kN.

Use grade 30 concrete with high-yield steel reinforcement to calculate:

6.1 The required number and diameter of the longitudinal bars (7)

6.2 The maximum and minimum percentage of the steel reinforcement (4)

6.3 The diameter and pitch of the helical binder (4)

[15]

QUESTION 7

FIGURE 5 shows a dimensioned T-beam.

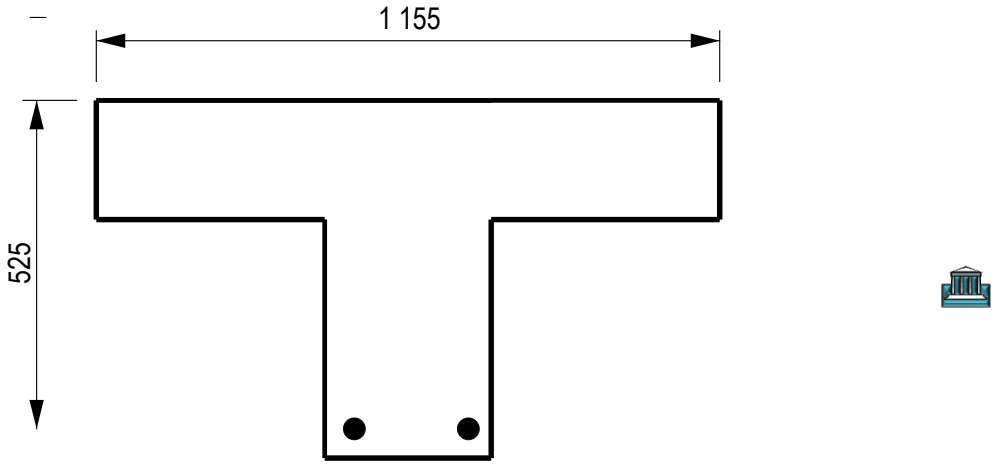


FIGURE 5

Calculate the required tension reinforcement if grade 25 concrete with high-yield tensile steel reinforcement is used. The ultimate bending moment to be resisted by the T-beam is 285 kNm. The neutral axis is within the flange of the flanged beam. **[9]**

TOTAL: 100

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SCHEDULE A

CROSS-SECTIONAL AREA OF REINFORCEMENT RODS FOR BEAMS AND COLUMNS										
Number of rods	Rod diameter (mm)									
	Ø6	Ø8	Ø10	Ø12	Ø16	Ø20	Ø25	Ø32	Ø40	Ø50
1	28,3	50,3	78,5	113,1	201,1	314,2	490,9	804,2	1 256,6	1 963,5
2	57	101	157	226	402	628	982	1 608	2 513	3 927
3	85	151	236	339	603	943	1 473	2 413	3 770	5 891
4	113	201	314	452	804	1 257	1 964	3 217	5 026	7 854
5	141	251	393	566	1 006	1 571	2 455	4 021	6 283	9 818
6	170	302	471	679	1 207	1 885	2 945	4 825	7 540	11 781
7	198	352	550	792	1 408	2 199	3 436	5 629	8 796	13 745
8	226	402	628	905	1 609	2 514	3 927	6 434	10 053	15 708
9	255	453	707	1 018	1 810	2 828	4 418	7 238	11 309	17 672
10	283	503	785	1 131	2 011	3 142	4 909	8 042	12 566	19 635
11	311	553	864	1 244	2 212	3 456	5 400	8 846	13 823	21 599
12	339	603	942	1 357	2 413	3 770	5 891	9 650	15 079	23 562
	Typical secondary reinforcement			Typical main reinforcement						

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SCHEDULE B

CROSS-SECTIONAL AREA OF REINFORCEMENT RODS PER METRE WIDTH FOR SLABS AND STAIRCASES										
Spacing of rods centre to centre	Rod diameter (mm)									
	Ø6	Ø8	Ø10	Ø12	Ø16	Ø20	Ø25	Ø32	Ø40	Ø50
50	566	1 005	1 571	2 262	4 021	6 283	9 817	16 085	25 133	39 270
75	377	670	1 048	1 508	2 681	4 189	6 545	10 723	16 755	26 180
100	283	503	785	1 131	2 011	3 142	4 909	8 042	12 566	19 635
125	226	402	628	905	1 608	2 513	3 927	6 434	10 053	15 708
150	188	335	524	754	1 340	2 094	3 272	5 362	8 378	13 090
175	162	387	449	646	1 149	1 795	2 805	4 596	7 181	11 220
200	141	251	393	565	1 005	1 571	2 454	4 021	6 283	9 817
250	113	201	314	452	804	1 257	1 963	3 217	5 027	7 854
300	94	168	262	377	670	1 047	1 636	2 681	4 189	6 545
350	81	144	224	323	574	898	1 402	2 298	3 590	5 610
400	71	125	196	283	503	786	1 227	2 011	3 142	4 909
500	57	101	157	226	402	620	982	1 608	2 513	3 927
	Typical secondary reinforcement				Typical main reinforcement					

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SCHEDULE C

ISOMETRIC BLACK HEXAGON BOLTS AND NUTS									
Normal size and thread diameter	DIMENSIONS IN MILLIMETRE								
	Pitch of thread coarse pitch series	Maximum width of head and nut		Maximum height of head		Maximum thickness of nut		Tensile stress area in mm²	Minimum distance between centres
		Across flats	Across corners	Black	Face on underside	Black	Face one side		
M 6	1	10,00	11,5	4,375	4,25	5,375	5	20,1	15
M 8	1,25	13,00	15,0	5,875	5,74	6,875	6,5	36,6	20
M 10	1,5	17,00	19,6	7,45	7,29	8,45	8,	58,0	25
M 12	1,75	19,00	21,9	8,45	8,29	10,45	10	84,3	30
M 16	2	24,00	27,7	10,45	10,29	13,45	13	157	40
M 20	2,5	30,00	34,6	13,90	13,35	16,55	16	245	50
M 22	2,5	32,00	36,9	14,90	14,35	18,55	18	303	55
M 24	3	35,00	41,6	15,90	15,35	19,65	19	353	60
M 27	3	41,00	47,3	17,90	17,35	22,65	22	459	67,5
M 30	3,5	46,00	53,1	20,05	19,42	24,65	24	561	75
M 33	3,5	50,00	57,7	22,05	21,42	26,65	26	694	82,5
M 36	4	55,00	63,5	25,05	23,42	29,65	29	817	90
M 39	4	60,00	69,3	26,05	25,42	31,80	31	976	97,5
M 42	4,5	65,00	75,1	27,05	26,42	34,80	34	1 120	105
M 45	4,5	70,00	80,8	29,05	28,42	36,80	36	1 300	112,5
M 48	5,0	75,00	86,6	31,05	30,42	38,80	38	1 470	120
M 52	5,0	80,00	92,4	34,25	33,42	42,80	42	1 760	130
M 56	5,5	85,00	98,1	36,25	35,50	45,80	45	2 030	140

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